

Worksheet 3A: FOIL and Expanding

Annotated Revision Guide

AIML Foundations Mathematics

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Part A: Single Bracket Expansion (Problems 1–8)

Concept. Expanding brackets applies the **distributive law**: $a(b + c) = ab + ac$. Every term inside the bracket gets multiplied by the term outside. With a negative outside factor, every sign inside flips.

Method. Multiply the outer term by each inner term in order. Collect like terms (same variable and same power) at the end. A common mnemonic: every term inside gets “visited exactly once.”

Watch out. $-(x + 3) \neq -x + 3$. The minus sign distributes to *every* term: $-(x + 3) = -x - 3$. Similarly, $2(x^2 + 3x - 1) = 2x^2 + 6x - 2$, not $2x^2 + 3x - 1$.

Where this leads. Every factoring problem in Part B of Worksheet 3B reverses expansion. If you can expand fluently, checking your factorisations by re-expanding becomes automatic.

Explore further. The distributive law is an *axiom* of ring theory in abstract algebra — the single rule that makes polynomial arithmetic work. For a beautiful algebraic structure built entirely on this, see Herstein, *Topics in Algebra*.

1. $3(x + 4)$

Distribute. $3x + 12$.

2. $-2(x - 5)$

Distribute. $-2x + 10$.

Note: Both signs flip: $-2 \times (-5) = +10$.

3. $x(x + 7)$

Distribute. $x^2 + 7x$.

4. $2x(3x - 4)$

Distribute. $6x^2 - 8x$.

5. $-3x(2x^2 - x + 1)$

Distribute. $-6x^3 + 3x^2 - 3x$.

Note: Three terms, each sign flips.

6. $4a^2(a - 3b)$

Distribute. $4a^3 - 12a^2b$.

7. $\frac{1}{2}(6x - 4)$

Distribute. $3x - 2$.

Note: Multiply numerators: $\frac{6x}{2} - \frac{4}{2}$.

8. $(x + 2)(x + 3)$ using the grid method

	$\times$	x	3
Grid.	x	x^2	$3x$
	2	$2x$	6

Sum. $x^2 + 3x + 2x + 6 = x^2 + 5x + 6$.

Note: The grid method is equivalent to FOIL but scales to any size bracket.

Part B: FOIL — Two Brackets (Problems 9–18)

Concept. **FOIL** is a mnemonic for multiplying two binomials: $(a + b)(c + d) = \underbrace{ac}_{\text{First}} + \underbrace{ad}_{\text{Outer}} + \underbrace{bc}_{\text{Inner}} + \underbrace{bd}_{\text{Last}}$. It is simply the distributive law applied twice: each term in the first bracket multiplies each term in the second.

Method. First $\times$ First; Outer $\times$ Outer; Inner $\times$ Inner; Last $\times$ Last. Then collect like terms (the outer and inner terms are usually the ones that combine). Alternatively, use the grid: a 2×2 table of products.

Watch out. Forgetting to collect like terms: $(x + 3)(x + 2) = x^2 + 2x + 3x + 6$, *not* $x^2 + 2x + 3x + 6$ left as four terms. Always combine $2x + 3x = 5x$ to get $x^2 + 5x + 6$.

Where this leads. Every quadratic you will ever factor came from FOILING two binomials. Factoring is FOIL in reverse. The special products (perfect square, difference of squares) below are the most useful patterns to memorise.

Explore further. FOIL generalises: three brackets need 8 terms (or a $2 \times 2 \times 2$ grid). The **binomial theorem** $\sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$ extends this to $(a + b)^n$ for any n — look up Pascal's triangle and see problem 20 below for a preview.

9. $(x + 2)(x + 5)$

FOIL. $x^2 + 5x + 2x + 10 = x^2 + 7x + 10$.

10. $(x - 3)(x + 4)$

FOIL. $x^2 + 4x - 3x - 12 = x^2 + x - 12$.

Note: Note: $+4x - 3x = +x$.

11. $(x - 6)(x - 1)$

FOIL. $x^2 - x - 6x + 6 = x^2 - 7x + 6$.

12. $(2x + 1)(x + 3)$

FOIL. $2x^2 + 6x + x + 3 = 2x^2 + 7x + 3$.

13. $(3x - 2)(2x + 5)$

FOIL. $6x^2 + 15x - 4x - 10 = 6x^2 + 11x - 10$.

14. $(x + 4)^2$

Recognise. Perfect square: $(a + b)^2 = a^2 + 2ab + b^2$.

Apply. $x^2 + 2(x)(4) + 16 = x^2 + 8x + 16$.

Note: Common mistake: $(x + 4)^2 \neq x^2 + 16$. The middle term $2 \cdot x \cdot 4 = 8x$ is essential.

15. $(x - 5)^2$

Perfect square. $x^2 - 10x + 25$.

16. $(x + 7)(x - 7)$

Difference of squares. $(a + b)(a - b) = a^2 - b^2$.

Apply. $x^2 - 49$.

Note: The middle terms cancel: $+7x - 7x = 0$. This pattern is crucial for Part C.

17. $(2x + 3)(2x - 3)$

Diff. of squares. $(2x)^2 - 9 = 4x^2 - 9$.

18. $(x + 1)(x^2 - x + 1)$

Distribute. $x(x^2 - x + 1) + 1(x^2 - x + 1)$.

Expand. $x^3 - x^2 + x + x^2 - x + 1 = x^3 + 1$.

Note: This is the sum of cubes: $(x+1)(x^2-x+1) = x^3+1$. Memorise it.

Part C: Special Products and Pascal's Triangle (Problems 19–25)

Concept. Three special products are worth memorising: (1) $(a+b)^2 = a^2+2ab+b^2$, (2) $(a-b)^2 = a^2-2ab+b^2$, (3) $(a+b)(a-b) = a^2-b^2$ (difference of squares). These patterns appear constantly in factoring, completing the square, and trigonometric identities.

Method. Recognise the pattern, apply the formula, double-check by expanding. For $(a+b)^n$ with larger n , use **Pascal's triangle** for the coefficients: row n of the triangle gives the binomial coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.

Watch out. $(a+b)^3 \neq a^3+b^3$. The correct expansion is $a^3+3a^2b+3ab^2+b^3$. The intermediate terms come from Pascal's row 3: 1, 3, 3, 1.

Where this leads. The difference of squares pattern $(a^2-b^2) = (a-b)(a+b)$ is the single most useful factoring pattern. It reappears in rationalising surds, partial fractions, and trigonometric proofs.

Explore further. Pascal's triangle encodes combinatorics: each entry $\binom{n}{k}$ counts the number of ways to choose k items from n . It appears in probability theory, the normal distribution (bell curve), and Sierpiński's triangle (fractal geometry).

19. $(a+b)^2$ — verify the formula

Expand by FOIL. $(a+b)(a+b) = a^2+ab+ab+b^2 = a^2+2ab+b^2$.

Memorise. $(a+b)^2 = a^2+2ab+b^2$

Note: The $2ab$ term is the one people most often forget.

20. Expand $(x+y)^3$ using Pascal's triangle

Row 3 of Pascal. Coefficients: 1, 3, 3, 1.

Apply. $x^3+3x^2y+3xy^2+y^3$.

Note: Powers of x decrease from 3 to 0; powers of y increase from 0 to 3.

21. $(3x-2y)^2$

Pattern. $(a-b)^2 = a^2-2ab+b^2$ with $a=3x$, $b=2y$.

Apply. $9x^2-12xy+4y^2$.

22. $(5x+2)(5x-2)$

Diff. of squares. $(5x)^2-4 = 25x^2-4$.

23. $(x^2+3)(x^2-3)$

Diff. of squares. $x^4 - 9$.